

ATUL SHARMA

Data Scientist

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PROFESSIONAL SUMMARY

Results-driven Data Scientist with 4 years of experience in Machine Learning, Deep Learning, and Generative AI solutions. Skilled in building scalable AI systems, enterprise-grade RAG applications, and multi-agent AI workflows using LangChain, LangGraph, GPT-4, Mistral, and FastAPI. Experienced in developing AI copilots, chatbot systems, and predictive models using XGBoost, Random Forest. Proficient in FAISS, Pinecone, Docker, Kubernetes, AWS, and cloud-native deployment architectures.

TECHNICAL SKILLS

Programming Languages: Python, SQL, MySQL, PostgreSQL

Machine Learning & Data Science: Scikit-learn, XGBoost, Logistic Regression, Random Forest, Feature Engineering, Model Evaluation, Cross Validation, Hyperparameter Tuning, Data Preprocessing, SMOTE, EDA, Predictive Analytics

Deep Learning & NLP: TensorFlow, Keras, PyTorch (Basic), CNN, RNN, LSTM, NLP, Text Classification

Generative AI & LLMs: GPT-4, Mistral 7B, LLaMA 3, BERT, Prompt Engineering, RAG, Agentic AI, Multi-Agent Systems, Conversational AI, Tool Calling, Function Calling

Agentic AI & Orchestration: LangGraph, LangChain Agents, Workflow Orchestration, Dynamic Tool Routing, Stateful AI Agents, Conversational Memory, Autonomous AI Workflows

RAG & Retrieval Systems: Semantic Search, Hybrid Retrieval, BM25, Cross-Encoder Reranking, Embedding Models, Recursive Chunking, Metadata Filtering, Vector Search

GenAI Frameworks & APIs: LangChain, LangGraph, Hugging Face Transformers, OpenAI API, Hugging Face Hub

Vector Databases & Caching: Pinecone, FAISS, Redis

Backend & APIs: FastAPI, Flask, REST APIs, Async Processing

Cloud & DevOps: AWS (EC2, S3, Lambda), Docker, Kubernetes, Cloud-Native Deployment

Monitoring & Observability: LangSmith, Prometheus, Grafana

Data Processing & Automation: Pandas, NumPy, OCR, Data Ingestion Pipelines

Tools & Platforms: Git, GitHub, VS Code, Jupyter Notebook, Google Colab, Streamlit

PROFESSIONAL EXPERIENCE

•Data Scientist | Orbex Technologies Pvt. Ltd| Delhi

Generative AI & Machine Learning Engineer

June 2022 – Present

- Developed enterprise AI Copilot platform using LangGraph and RAG architecture.
- Built hybrid retrieval systems using vector search, BM25, and reranking.
- Developed NL-to-SQL workflows for conversational enterprise analytics.
- Implemented Redis conversational memory and FastAPI backend services.
- Worked on Docker, Kubernetes, and AWS deployment for scalable AI infrastructure.
- Collaborated on AI integration, API development, deployment optimization, and monitoring.

PROJECTS

•Enterprise Multi-Agent AI Copilot Platform

GenAI – Agentic AI Enterprise Solution

- Developed an enterprise-grade AI Copilot platform enabling intelligent interaction with structured (SQL) and unstructured enterprise data using natural language queries.
- Designed and implemented multi-agent AI workflows using LangGraph for intelligent routing, orchestration, and dynamic tool selection.
- Built advanced RAG pipelines including document ingestion, chunking, embedding generation, semantic retrieval, metadata filtering, and contextual response generation.
- Implemented hybrid retrieval architecture using Vector Search, BM25 retrieval, and reranking techniques to improve retrieval precision and reduce hallucinations.
- Developed Natural Language-to-SQL workflows enabling non-technical users to query enterprise databases conversationally.
- Built scalable FastAPI backend services, Redis conversational memory, and multimodal AI workflows with OCR-based document understanding.
- Implemented monitoring and observability pipelines using LangSmith, Prometheus, and Grafana for tracking retrieval quality, latency, hallucination rate, and agent performance.
- Collaborated on Dockerization, Kubernetes orchestration, AWS deployment workflows, and scalable cloud-native infrastructure optimization.
- Technologies Used: Python, FastAPI, LangChain, LangGraph, GPT-4o, Pinecone, FAISS, Redis, PostgreSQL, Docker, Kubernetes, AWS, Streamlit, OCR, Prometheus, Grafana, LangSmith.

•AI-Powered Customer Support Chatbot

RAG-based intelligent chatbot using LLMs and semantic search

- Built chatbot using LangChain and FAISS for intelligent query handling.
- Implemented semantic retrieval and contextual response generation using LLMs.
- Developed backend APIs using FastAPI and frontend using Streamlit.
- Reduced customer support workload by nearly 50% through automated AI-driven responses.
- Technologies Used: Python, FAISS, FastAPI, Streamlit, GPT/Mistral.

•Customer Churn Prediction System

Machine Learning system for churn prediction and retention analytics

- Built and compared Logistic Regression, Random Forest, and XGBoost models for churn prediction.
- Achieved 0.90 ROC-AUC using XGBoost with hyperparameter tuning and feature engineering.
- Used SMOTE for handling imbalanced datasets and improving prediction stability.
- Developed FastAPI APIs for real-time churn prediction and business analytics integration.
- Technologies Used: Python, Scikit-learn, XGBoost, SHAP, Docker.

EDUCATION

•Bachelor of Computer Applications (BCA)

Kalinga University

2020– 2023